

Supplementary File S7

RQ7: Challenges, Future Directions, and Evolving Priorities

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ7. It reports dominant challenge families, future research orientations, cross-survey shift signals, and concise survey-level answers.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

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S7.1 A review of reinforcement learning for autonomous building energy management [1]

Table S7.1: RQ7 framing and shift coding for Mason and Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper primarily frames the field’s problems around algorithmic capability, scaling, multi-objective control, transfer, and robustness.
Secondary challenge family	Operational / deployment	Simulation dependence and costly physical deployment are strongly present through the paper’s discussion of the difficulty of direct trial-and-error learning in real buildings.
Safety / trust emphasis	Limited	Safety appears only indirectly through the risk and cost of trial-and-error learning in real deployment; explicit trust, privacy, explainability, ethics, and governance are absent.
Dominant future orientation	Algorithmic advancement	The future agenda is led by deep reinforcement learning, multi-task learning, transfer learning, multi-objective reinforcement learning, meta-learning, and multi-agent reinforcement learning.
Shift signal: simulation → real-world deployment	1	The paper clearly highlights dependence on simulation and the difficulty of physical deployment.
Shift signal: single-home → community / larger-scale coordination	1	The paper points toward communities of homes and the scalability of multi-agent reinforcement learning.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded.
Shift signal: performance → safety / trust / governance	0	No explicit move in that direction is developed.
Overall RQ7 contribution strength	Strong	The paper contains an explicit limitations and future-directions section, so its RQ7 contribution is structurally visible even though the framing remains mostly technical.

Table S7.2: RQ7 challenge and future-direction scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Mason and Grijalva (2019)	3	3	2	2	3	3	0	1	0	3	2	2	2	2	3	3	3	0	0	0

Table S7.3: RQ7 summary table for Mason and Grijalva (2019), after recoding.

Item	Summary
How the paper answers RQ7	Mason and Grijalva (2019) answer RQ7 mainly by identifying technical and deployment-related obstacles to reinforcement learning adoption in building energy management and by proposing a forward-looking algorithmic agenda for overcoming them.
Most emphasized challenges	Simulation dependence, costly trial-and-error learning in real buildings, multi-objective trade-offs, generalization and transfer difficulty, scalability to communities of homes, and uncertainty about policy robustness under major environmental change.
Most emphasized future directions	Deep reinforcement learning, transfer learning, multi-task learning, multi-objective reinforcement learning, meta-learning, multi-agent reinforcement learning, robustness to environmental change, and pathways from simulation to physical deployment.
Barrier distribution	Mostly technical; strong secondary operational/deployment emphasis; minimal explicit safety, trust, governance, privacy, or ethics framing.
Temporal outlook	Early-stage review in which priorities remain centered on improving reinforcement-learning capability, scalability, transfer, and robustness rather than on governance, explainability, ethics, or organizational adoption.
Usefulness for cross-survey synthesis	High for identifying early technical priorities and the early emergence of simulation-to-real, transfer, scalability, and robustness concerns.

Table S7.4: One-line RQ7 answer for Mason and Grijalva (2019), after recoding.

One-line RQ7 answer
Mason and Grijalva (2019) frame the main barriers to reinforcement-learning-based building energy management primarily as technical and simulation/deployment-related, especially simulation dependence, costly real-building training, multi-objective complexity, transfer and scalability challenges, and uncertain robustness under major environmental change, while pointing future work toward deep reinforcement learning, transfer and multi-task learning, multi-objective methods, meta-learning, and multi-agent control at larger scales.

S7.2 Reinforcement learning for building controls: The opportunities and challenges [2]

Table S7.5: RQ7 framing and shift coding for Wang and Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper foregrounds training burden, generalization, robustness, safety/security, and multi-agent coordination as major obstacles.
Secondary challenge family	Operational / deployment	Limited real-building adoption, simulation dominance, and benchmarking difficulties are strongly present as deployment-facing barriers.
Safety / trust emphasis	Moderate–strong	Safety, control security, and robustness are structurally discussed, although broader governance, privacy, ethics, and explainability are not developed.
Dominant future orientation	Deployment-ready algorithmic improvement	The future agenda aims to make reinforcement learning more practical for real buildings through faster training, stronger robustness, transfer learning, multi-agent coordination, and shared evaluation platforms.
Shift signal: simulation → real-world deployment	1	The paper explicitly highlights limited real-building implementations and argues for progress toward real deployment.
Shift signal: single-building → community / larger-scale coordination	1	The paper identifies multi-agent control for building and campus scale as an important challenge.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded.
Shift signal: performance → safety / trust / governance	1	The paper clearly elevates safety, security, and robustness, although not governance or ethics.
Overall RQ7 contribution strength	Strong	The paper directly organizes the field around key challenges and future opportunities, including training burden, robustness, generalization, transfer, multi-agent control, and benchmarking.

Table S7.6: RQ7 challenge and future-direction scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Wang and Hong (2020)	3	3	2	1	3	3	0	3	0	0	0	3	0	0	3	3	3	0	3	0

Table S7.7: RQ7 summary table for Wang and Hong (2020), after recoding.

Item	Summary
How the paper answers RQ7	Wang and Hong (2020) answer RQ7 by identifying the main barriers to wider reinforcement learning adoption in building controls and by outlining a future agenda focused on making reinforcement learning more practical, robust, transferable, safe, and comparable across studies.
Most emphasized challenges	Limited real-building adoption, simulation dependence, time- and data-intensive training, safety/security/robustness concerns, weak generalization, and multi-agent coordination complexity.
Most emphasized future directions	Transfer learning, faster training, stronger robustness and safety, multi-agent reinforcement learning for larger-scale coordination, real-world deployment pathways, and open-source testbeds/datasets for performance benchmarking.
Barrier distribution	Mostly technical; strong secondary operational/deployment emphasis; strong safety/robustness emphasis; minimal trust, privacy, ethics, or governance framing.
Temporal outlook	Early review that moves beyond pure algorithm cataloguing toward a more deployment-aware framing, especially around generalization, safety, robustness, and evaluation infrastructure.
Usefulness for cross-survey synthesis	Very high for tracing the early consolidation of training burden, robustness, transfer learning, simulation-to-real deployment, multi-agent control, and benchmarking as recurring priorities in the field.

Table S7.8: One-line RQ7 answer for Wang and Hong (2020), after recoding.

One-line RQ7 answer
Wang and Hong (2020) frame the main barriers to reinforcement-learning-based building controls as limited real-building deployment, costly and data-intensive training, weak robustness/security, limited generalization, and multi-agent coordination challenges, and they point future work toward transfer learning, faster training, stronger safety and robustness, real-world deployment, and open-source benchmarking infrastructure to support practical adoption.

S7.3 The reinforcement learning method for occupant behavior in building control: A review [3]

Table S7.9: RQ7 framing and shift coding for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around occupant stochasticity, transfer difficulty, rigorous training requirements, high-dimensional state/action design, and the underdeveloped status of occupant-aware reinforcement learning methods.
Secondary challenge family	Operational / implementation	A secondary concern is the practical difficulty of repeatedly collecting occupant behavior data and feedback during implementation.
Safety / trust emphasis	Limited	Control security and robustness are mentioned, but broader trust, governance, privacy, ethics, and explainability themes are absent.
Dominant future orientation	Occupant-aware technical development	The future agenda is led by deeper reinforcement learning methods, better approximation for complex problems, richer occupant feedback integration, and broader control applications beyond the current HVAC-heavy focus.
Shift signal: simulation → real-world deployment	0	No strong, explicit shift of this kind is foregrounded as a central narrative.
Shift signal: single-home → community / larger-scale coordination	0	This is not a central future orientation in the paper.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	The paper centers occupant interaction, occupant feedback, stochastic behavior, and individual differences as key parts of the control problem.
Shift signal: performance → safety / trust / governance	0	No explicit move in that direction is developed.
Overall RQ7 contribution strength	Strong	The whole review is organized around occupant-aware reinforcement learning challenges and future research needs, even though it is less expansive on deployment, infrastructure, safety, or governance.

Table S7.10: RQ7 challenge and future-direction scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Han et al. (2021)	1	3	3	1	3	0	0	1	0	3	0	2	0	0	0	2	1	3	0	0

Table S7.11: RQ7 summary table for Han et al. (2021), after recoding.

Item	Summary
RQ7 answer	Han et al. (2021) answer RQ7 by identifying the main difficulties of occupant-aware reinforcement learning control in buildings and by pointing toward technically richer ways of representing, learning, and using occupant behavior in future control systems.
Key challenges	Occupant stochasticity and uncertainty, poor transferability across settings, rigorous training requirements, high-dimensional state/action design, unclear occupant-agent interaction, and the still underdeveloped status of occupant-aware reinforcement learning methods.
Key future directions	Deep reinforcement learning for complex problems, better approximation methods for high-dimensional and continuous control, transfer-aware modeling, richer integration of occupant feedback and interaction, broader applications beyond HVAC, and stronger development of the field through wider collaboration and attention.
Overall RQ7 profile	Mostly technical and occupant-centered, with a weaker implementation layer and only limited safety framing; especially useful for showing how future directions were still being formulated around modeling, learning, and integrating occupant-related control behavior.

Table S7.12: One-line RQ7 answer for Han et al. (2021), after recoding.

One-line RQ7 answer
Han et al. (2021) frame the main challenges of reinforcement learning for occupant behavior in building control as occupant stochasticity, poor transferability, rigorous training, and high-dimensional representation problems, and they point future work toward deep reinforcement learning, better approximation for complex continuous problems, richer occupant-feedback integration, and broader occupant-aware control applications beyond the current HVAC-dominant focus.

S7.4 A review of deep reinforcement learning for smart building energy management [4]

Table S7.13: RQ7 framing and shift coding for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around modeling difficulty, uncertainty, coupled constraints, high-dimensional and hybrid decision complexity, data inefficiency, scalability, transferability, and multi-objective smart-building energy management.
Secondary challenge family	Operational / implementation	A secondary concern is the practical difficulty of collecting enough data, reducing exploration cost, and making DRL methods feasible when real-building interaction and historical data are limited.
Safety / trust emphasis	Limited	The paper notes that trial-and-error exploration can cause discomfort and high energy consumption, but broader trust, governance, privacy, ethics, or verified safety themes are absent.
Dominant future orientation	Scalable and data-efficient DRL development	The future agenda is led by offline and model-based DRL, transfer learning, meta-RL, hierarchical DRL for multi-timescale control, advanced multi-objective DRL, scalable multi-agent methods, and DRL for building microgrids.
Shift signal: simulation → real-world deployment	1	The paper partially signals this shift by noting that many model-free DRL methods remain impractical because of data and exploration burdens, while offline and model-based pathways may make implementation more feasible.
Shift signal: single-building → community / larger-scale coordination	1	The paper expands the discussion from single subsystems to multiple subsystems, multi-zone buildings, multi-building optimization, multi-agent control, and building microgrids.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded as a central narrative.
Shift signal: performance → safety / trust / governance	0	No explicit move toward safety, trust, governance, privacy, or ethics is developed.
Overall RQ7 contribution strength	Strong	The paper directly organizes the field around unresolved challenges and future research directions for DRL-based smart-building energy management.

Table S7.14: RQ7 challenge and future-direction scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Yu et al. (2021)	2	3	3	3	3	3	0	1	0	2	0	3	3	3	3	2	3	0	0	0

Table S7.15: RQ7 summary table for Yu et al. (2021), after recoding.

Item	Summary
RQ7 answer	Yu et al. (2021) answer RQ7 by identifying the main technical and practical difficulties of DRL-based smart-building energy management and by outlining a future agenda aimed at making DRL more data-efficient, scalable, transferable, and capable of handling complex multi-objective building-energy problems.
Key challenges	Difficult building modeling, uncertain parameters, spatially and temporally coupled constraints, large and hybrid decision spaces, costly data collection and trial-and-error exploration, conflicting objectives, multi-zone and multi-building scalability limits, and weak transferability when historical data are scarce.
Key future directions	Offline and model-based DRL, transfer learning, meta-RL, hierarchical DRL for multi-timescale optimization, advanced multi-objective DRL, scalable multi-agent methods for multi-zone and multi-building settings, and DRL-based optimization of building microgrids.
Overall RQ7 profile	Mostly technical, with a clear practical data-efficiency and implementation layer and only limited safety framing; especially useful for showing how the field's future agenda broadened toward hierarchy, scalability, transfer, multi-objective control, and microgrid coordination.

Table S7.16: One-line RQ7 answer for Yu et al. (2021), after recoding.

One-line RQ7 answer
Yu et al. (2021) frame the main challenges of DRL-based smart-building energy management as modeling difficulty, uncertainty, coupled and large-scale decision problems, costly exploration, conflicting objectives, and weak scalability and transferability, while pointing future work toward offline and model-based DRL, transfer learning, meta-RL, hierarchical multi-timescale control, advanced multi-objective DRL, scalable multi-agent methods, and DRL-based optimization of building microgrids.

S7.5 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [5]

Table S7.17: RQ7 framing and shift coding for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around action-space design, abstraction level, indoor-environment representation, actuator/setpoint mapping, and computational scalability in multi-zone HVAC control.
Secondary challenge family	Operational / implementation	A secondary concern is the weakly specified linkage between RL controllers and real HVAC or building-automation architectures such as BACS, HEMS, BEMS, and VPP contexts.
Safety / trust emphasis	Limited	The paper is concerned with physically meaningful and implementable control formulations, but broader safety, trust, privacy, ethics, or governance themes are not developed.
Dominant future orientation	Deployment-compatible HVAC-aware formulation	The future agenda is led by clearer RL-to-HVAC/BACS mappings, more realistic and less over-abstract formulations, stronger linkage to HEMS/BEMS/VPP systems, and more scalable multi-zone control formulations.
Shift signal: simulation → real-world deployment	1	The paper partially signals this shift by identifying the literature gap around how RL agents should be connected to physical building systems and deployable HVAC control layers.
Shift signal: single-building → community / larger-scale coordination	0	This is not a main organizing shift in the paper, although multi-zone scalability is central.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded as a central narrative.
Shift signal: performance → safety / trust / governance	0	No explicit move in that direction is developed.
Overall RQ7 contribution strength	Strong	The paper directly identifies a concrete literature gap, specifies the main technical and implementation challenges, and states clear directions for further research on deployable HVAC-aware RL formulations.

Table S7.18: RQ7 challenge and future-direction scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Sierla et al. (2022)	0	0	3	1	1	2	3	1	0	0	0	0	0	0	2	0	3	1	0	0

Table S7.19: RQ7 summary table for Sierla et al. (2022), after recoding.

Item	Summary
RQ7 answer	Sierla et al. (2022) answer RQ7 by identifying the main challenges of RL-based HVAC control as weakly specified and overly abstract action formulations, inadequate representation of indoor-environment objectives, and poor scalability in multi-zone settings, while pointing future work toward more explicit, deployment-compatible, and HVAC-aware RL formulations.
Key challenges	Poorly specified action-to-system mappings, excessive abstraction that can hide important HVAC characteristics, narrow treatment of indoor-environment objectives, exponential action-space growth in multi-zone settings, and weak linkage to HEMS/BEMS/VPP and physical HVAC control contexts.
Key future directions	Clearer mapping of RL actions to actuator commands, setpoint adjustment, supervisory control, or planning layers; stronger linkage to HEMS/BEMS/VPP systems; more realistic HVAC-aware formulations; and more scalable approaches such as multi-agent RL for multi-zone buildings.
Overall RQ7 profile	Mostly technical with a clear implementation layer and little explicit safety/trust framing; especially useful for showing that future progress depends not only on better algorithms but also on better problem formulation and clearer connection to real building-control architectures.

Table S7.20: One-line RQ7 answer for Sierla et al. (2022), after recoding.

One-line RQ7 answer
Sierla et al. (2022) frame the main challenges of RL-based HVAC control as overly abstract and weakly specified action formulations, narrow representation of indoor-environment objectives, exponential multi-zone action-space growth, and weak linkage to real building-control architectures, and they point future work toward clearer HVAC/BACS mappings, stronger HEMS/BEMS/VPP linkage, and more scalable approaches such as multi-agent RL.

S7.6 Applications of reinforcement learning for building energy efficiency control: A review [6]

Table S7.21: RQ7 framing and shift coding for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around costly trial-and-error learning, long training time, simulation-to-real differences, security risk, and the difficulty of matching RL methods to increasingly complex building-control problems.
Secondary challenge family	Operational / implementation	A secondary concern is the real-world cost and risk of deploying RL in buildings, including inconvenience to occupants, potential equipment damage, uncontrollable real costs, and unsafe online exploration.
Safety / trust emphasis	Moderate–strong	The paper explicitly mentions security problems and undesirable real-world effects of trial-and-error control, although broader governance, privacy, ethics, explainability, and trust themes are absent.
Dominant future orientation	Advanced RL for complex and occupant-aware building control	The future agenda is led by DRL, MARL, faster training through prior knowledge, and stronger integration of occupant-related building control.
Shift signal: simulation → real-world deployment	1	The paper explicitly discusses the high real cost of direct RL deployment and the differences between simulated and real environments.
Shift signal: single-building → community / larger-scale coordination	1	MARL is presented as a future direction for complex multi-object control problems, including multi-building and smart-grid-related settings.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	The paper explicitly highlights occupant-related building control as an important future direction.
Shift signal: performance → safety / trust / governance	0	No clear shift of this kind is developed beyond practical caution about security and real-world risk.
Overall RQ7 contribution strength	Strong	The paper explicitly separates limitations and future directions and directly frames them as one of the review’s main contributions.

Table S7.22: RQ7 challenge and future-direction scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Fu et al. (2022)	2	3	2	0	2	3	0	3	0	3	0	0	0	0	3	0	2	3	2	0

Table S7.23: RQ7 summary table for Fu et al. (2022), after recoding.

Item	Summary
RQ7 answer	Fu et al. (2022) answer RQ7 by identifying the main practical and technical difficulties of applying reinforcement learning to building energy-efficiency control and by outlining a future agenda aimed at making RL better suited to increasingly complex, multi-agent, and occupant-aware building-control problems.
Key challenges	Costly and risky trial-and-error in real buildings, long training time, differences between simulated and real environments, security risk, occupant inconvenience, possible equipment damage, and difficulty adapting RL methods to diverse and increasingly complex building-control settings.
Key future directions	Deep reinforcement learning, multi-agent reinforcement learning, reducing training time through prior knowledge, stronger simulation-to-real awareness, and stronger integration of occupant-related building control.
Overall RQ7 profile	Mostly technical, with a clear implementation layer and moderate-to-strong safety/security framing; especially useful for showing a shift toward DRL, MARL, faster training, and occupant-related control as future priorities in building energy-efficiency research.

Table S7.24: One-line RQ7 answer for Fu et al. (2022), after recoding.

One-line RQ7 answer
Fu et al. (2022) frame the main challenges of reinforcement-learning-based building energy-efficiency control as costly and risky trial-and-error in real buildings, long training time, simulation-to-real differences, security risks, and difficulty matching RL methods to increasingly complex building-control problems, while pointing future work toward DRL, MARL, faster training with prior knowledge, and stronger integration of occupant-related control.

S7.7 Systematic review on deep reinforcement learning-based energy management for different building types [7]

Table S7.25: RQ7 framing and shift coding for Shaqour and Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around data hunger, real-time modeling difficulty, multiple objectives, generalization and scaling limits, and the heavy data requirements of DRL-based BEMS.
Secondary challenge family	Operational / implementation	A strong secondary concern is the lack of real implementations and system validations, with only a small minority of recent studies including real-world validation.
Safety / trust emphasis	Limited–moderate	Comfort, health, and acceptable deployment behavior are discussed, but broader trust, explainability, governance, privacy, or ethics themes are not developed.
Dominant future orientation	Data-efficient and large-scale DRL development	The future agenda is led by transfer learning, federated learning, data-efficient DRL, more experimental and real-system validation, multi-agent district/grid-level scaling, and broader integration of renewables, storage, and demand response across building types.
Shift signal: simulation → real-world deployment	1	The paper explicitly highlights the lack of real implementations and calls for more experimental and real-system validation.
Shift signal: single-building → community / larger-scale coordination	1	The paper repeatedly points toward residential districts, community-level virtual power plants, multi-building optimization, and grid-level integration.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	The paper explicitly emphasizes thermal discomfort in offices, health aspects, and the direct user impact of demand response in residential buildings.
Shift signal: performance → safety / trust / governance	0	No clear move in that direction is developed beyond practical concern for comfort and acceptable deployment.
Overall RQ7 contribution strength	Strong	The discussion section is organized explicitly around challenges and building-type-specific future research recommendations.

Table S7.26: RQ7 challenge and future-direction scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Shaqour and Hagishima (2022)	3	3	2	3	3	3	0	2	0	2	0	3	2	0	3	2	3	3	2	0

Table S7.27: RQ7 summary table for Shaqour and Hagishima (2022), after recoding.

Item	Summary
RQ7 answer	Shaqour and Hagishima (2022) answer RQ7 by identifying the main technical and practical difficulties of DRL-based building energy management across different building types and by outlining a future agenda aimed at making DRL more data-efficient, more scalable, and more broadly validated in real systems.
Key challenges	Heavy data requirements, lack of real implementations and validations, real-time modeling difficulty, multiple competing objectives, weak generalization and scaling, and building-type-specific integration gaps such as limited renewable, storage, and demand-response integration in some contexts.
Key future directions	Transfer learning, federated learning, data-efficient DRL, more experimental and real-system validation, multi-agent district- and grid-level optimization, stronger integration of renewables, storage, and demand response, more explicit attention to thermal discomfort and health-related objectives, and broader open-source or standardized research support.
Overall RQ7 profile	Mostly technical with a strong implementation layer and a noticeable human-centered strand; especially useful for showing how future directions diversify by building type while converging on data efficiency, scaling, and real validation as common priorities.

Table S7.28: One-line RQ7 answer for Shaqour and Hagishima (2022), after recoding.

One-line RQ7 answer
Shaqour and Hagishima (2022) frame the main challenges of DRL-based building energy management as data hunger, limited real validation, real-time modeling and multi-objective complexity, and weak large-scale generalization, while pointing future work toward transfer and federated learning, data-efficient DRL, more experimental deployment, multi-agent district- and grid-level integration, and stronger occupant- and health-aware control across building types.

S7.8 Building energy management with reinforcement learning and model predictive control: A survey [8]

Table S7.29: RQ7 framing and shift coding for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.		
Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main RL-related obstacles around low data efficiency, long training periods, safety under trial-and-error learning, robust adaptability in dynamic environments, and computational complexity.
Secondary challenge family	Operational / implementation	A secondary concern is the practical need for more reliable and implementable RL-based BEMS control schemes, especially in real-time and dynamically changing settings.
Safety / trust emphasis	Moderate–strong	Safety is explicitly highlighted as a core RL challenge, but broader trust, governance, privacy, ethics, and explainability themes are not developed.
Dominant future orientation	Safer and more data-efficient RL through hybridization	The future agenda is led by hybrid MPC–RL and model-based RL approaches aimed at improving data efficiency, safety, robustness, and practical implementation.
Shift signal: simulation → real-world deployment	0	No strong central narrative of this kind is foregrounded; the emphasis is more on method choice and RL improvement than on deployment evidence.
Shift signal: single-building → community / larger-scale coordination	0	This is not a main organizing shift in the paper.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded as a central narrative.
Shift signal: performance → safety / trust / governance	1	Safety is explicitly elevated as a key RL challenge, although broader trust, governance, privacy, or ethics issues are not addressed.
Overall RQ7 contribution strength	Strong	The paper explicitly names RL-side challenge categories and future hybrid directions, especially data efficiency, safety, robust adaptability, and MPC/RL or model-based reinforcement-learning integration.

Table S7.30: RQ7 challenge and future-direction scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.																				
Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Zhang et al. (2022)	1	3	1	1	3	0	0	3	0	0	0	0	0	0	0	3	2	0	3	0

Table S7.31: RQ7 summary table for Zhang et al. (2022), after recoding.

Item	Summary
RQ7 answer	Zhang et al. (2022) answer RQ7 by identifying the main RL-related difficulties in building energy management and by outlining a future agenda aimed at making RL-based control more data-efficient, safer, more robust, and easier to implement.
Key challenges	Low data efficiency, long training periods, safety under trial-and-error learning, limited robust adaptability in dynamic environments, computational complexity, and the practical difficulty of making RL reliable for BEMS control.
Key future directions	Hybrid MPC–RL and model-based RL approaches, safer learning schemes, robustness-oriented method design, and more data-efficient RL methods that improve reliability and practical implementation in BEMS.
Overall RQ7 profile	Mostly technical with a moderate implementation layer and a clear safety emphasis; especially useful for showing how RL-oriented future work is framed around sample efficiency, safe learning, robust adaptability, and hybrid control design.

Table S7.32: One-line RQ7 answer for Zhang et al. (2022), after recoding.

One-line RQ7 answer
Zhang et al. (2022) frame the main RL-related challenges in building energy management as low data efficiency, long training periods, safety under trial-and-error learning, and limited robust adaptability, while pointing future work toward hybrid MPC–RL and model-based RL approaches that improve data efficiency, safety, robustness, and practical implementation.

S7.9 Reinforcement learning: theory and applications in HEMS [9]

Table S7.33: RQ7 framing and shift coding for Al-Ani and Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main issues around the immaturity of RL in HEMS, especially the prevalence of simplified tabular and discrete formulations and the need for broader use of DNN-based reinforcement-learning methods for more complex HEMS scenarios.
Secondary challenge family	Operational / implementation	A secondary concern is the still very limited real-world deployment of RL controllers, with most studies remaining simulation-based.
Safety / trust emphasis	Limited	The paper does not build a broad safety, trust, privacy, ethics, explainability, or governance agenda.
Dominant future orientation	Methodological maturation of RL in HEMS	The future agenda is oriented toward broader use of DNN-based RL, better alignment of RL algorithm classes with HEMS domains, gradual expansion of multi-agent approaches, and more disciplined performance evaluation.
Shift signal: simulation → real-world deployment	1	The paper notes that only a small fraction of surveyed work was deployed in the real world, highlighting a maturity gap.
Shift signal: single-building → community / larger-scale coordination	0	This is not a main organizing shift in the paper, although multi-agent schemes are noted as an encouraging direction.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded as a central narrative.
Shift signal: performance → safety / trust / governance	0	No explicit move in that direction is developed.
Overall RQ7 contribution strength	Moderate–strong	The paper is stronger than a purely descriptive survey because it diagnoses field immaturity, simplified problem setups, low real-world deployment, and proposes concrete RL evaluation metrics, but it is less challenge/future-direction-structured than the strongest RQ7 papers.

Table S7.34: RQ7 challenge and future-direction scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Al-Ani and Das (2022)	2	0	2	1	0	1	0	0	0	3	0	0	0	0	2	0	1	0	0	0

Table S7.35: RQ7 summary table for Al-Ani and Das (2022), after recoding.

Item	Summary
RQ7 answer	Al-Ani and Das (2022) answer RQ7 by portraying RL in HEMS as a still-maturing field and by emphasizing that much of the literature remains based on simplified problem formulations, limited real-world deployment, and uneven movement toward richer DRL methods.
Key challenges	Very limited real-world deployment, frequent use of simplified tabular RL methods, widespread reliance on discrete state/action formulations, limited multi-agent maturity, and the resulting immaturity of the field for complex HEMS scenarios.
Key future directions	Broader use of DNN-based RL methods, gradual expansion of multi-agent approaches, and more systematic evaluation through explicit RL performance metrics.
Overall RQ7 profile	Mostly technical and methodological, with a secondary implementation-maturity layer; especially useful for showing that a major early priority in HEMS-focused RL research was moving beyond simplified tabular setups toward richer DRL methods and stronger evaluation practice.

Table S7.36: One-line RQ7 answer for Al-Ani and Das (2022), after recoding.

One-line RQ7 answer
Al-Ani and Das (2022) frame the main challenges of RL in HEMS as the immaturity of the field, the prevalence of simplified tabular and discrete formulations, and the still very limited real-world deployment of RL controllers, while pointing future work toward broader DNN-based RL use, some expansion of multi-agent schemes, and more systematic evaluation through explicit performance metrics.

S7.10 A review of reinforcement learning for controlling building energy systems from a computer science perspective [10]

Table S7.37: RQ7 framing and shift coding for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around sample efficiency, long training and data-collection times, transfer across buildings, multi-agent scaling, interoperability, explainability, and the shortage of theory for reinforcement learning in building energy system control.
Secondary challenge family	Operational / implementation	A secondary concern is the practical design of scalable data-processing and control infrastructures, including cloud-based computation, sensor–actuator communication, interoperability, and asynchronous multi-building data streams.
Safety / trust emphasis	Moderate	Safety is present through the risk of suboptimal exploratory control making buildings temporarily uninhabitable, while explainability and privacy are treated as important practical and ethical requirements.
Dominant future orientation	Sample-efficient and scalable reinforcement learning for building energy systems	The future agenda is led by model-based and sample-efficient reinforcement learning, transfer learning between buildings, multi-agent approaches for building clusters, explainability, stronger theory, and scalable computation infrastructures.
Shift signal: simulation → real-world deployment	0	No strong central deployment-shift narrative is foregrounded; the emphasis is more on computer-science bottlenecks to scaling reinforcement learning than on validation stages.
Shift signal: single-building → community / larger-scale coordination	1	The paper explicitly structures one challenge category around building clusters and another around multi-agent aspects.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded as a central narrative.
Shift signal: performance → safety / trust / governance	1	The paper elevates safety, explainability, and privacy as practical requirements, although it does not develop a full governance agenda.
Overall RQ7 contribution strength	Strong	The review is explicitly organized around computer-science-related challenge categories and future research directions rather than only around applications or algorithms.

Table S7.38: RQ7 challenge and future-direction scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Weinberg et al. (2023)	1	3	2	0	3	3	3	2	3	0	0	3	0	0	3	1	2	0	1	3

Table S7.39: RQ7 summary table for Weinberg et al. (2023), after recoding.

Item	Summary
RQ7 answer	Weinberg et al. (2023) answer RQ7 by identifying the main challenges of reinforcement-learning-based building energy control from a computer-science perspective and by outlining a future agenda aimed at making reinforcement learning more sample-efficient, transferable, explainable, scalable, and theoretically grounded in building energy systems.
Key challenges	Slow data collection and long training times, low sample efficiency, complex building dynamics, the difficulty of transferring policies and models between buildings, multi-agent scaling problems, practical interoperability issues, explainability and privacy requirements, and limited theoretical understanding of reinforcement learning for building energy system control.
Key future directions	Model-based and sample-efficient reinforcement learning, transfer learning between buildings, multi-agent reinforcement learning for building clusters, explainability and privacy-aware methods, stronger theoretical analysis, and scalable system-design and cloud/data-pipeline infrastructures.
Overall RQ7 profile	Mostly technical with a clear implementation layer and a strong computer-science challenge framing; especially useful for showing a shift toward sample efficiency, transfer, explainability, privacy, scalable system design, and multi-agent coordination as major future priorities.

Table S7.40: One-line RQ7 answer for Weinberg et al. (2023), after recoding.

One-line RQ7 answer
Weinberg et al. (2023) frame the main challenges of reinforcement-learning-based building energy control as sample inefficiency, long training and data-collection times, transfer across buildings, multi-agent scaling, interoperability, explainability, privacy, and scalable system design, while pointing future work toward model-based and sample-efficient reinforcement learning, transfer learning, multi-agent approaches, stronger theory, explainable reinforcement learning, and scalable infrastructures for large-scale building energy system control.

S7.11 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [11]

Table S7.41: RQ7 framing and shift coding for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.		
Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around weak generalization to real-world variability, insufficient diversification of exogenous training conditions, unsafe direct transfer to real buildings, and the resulting need for repeated retraining or fine-tuning.
Secondary challenge family	Operational / implementation	A secondary concern is the limited deployment of reinforcement learning in real buildings and the practical cost of adapting trained agents before real use.
Safety / trust emphasis	Moderate–strong	Safety is present through the concern that current training methods can prevent safe real-building use, although broader trust, governance, privacy, ethics, or explainability themes are not developed.
Dominant future orientation	Generalization-ready reinforcement learning for real buildings	The future agenda is led by improving adaptation to real-world variability through better training design and especially meta-reinforcement learning, so that reinforcement learning can be used more safely and efficiently in real buildings.
Shift signal: simulation → real-world deployment	1	The paper explicitly highlights the limited use of reinforcement learning in real buildings and argues that future work should focus on solving barriers to real deployment.
Shift signal: single-building → community / larger-scale coordination	0	This is not a main organizing shift in the paper.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	0	No clear shift of this kind is foregrounded as a central narrative.
Shift signal: performance → safety / trust / governance	1	The paper explicitly stresses safer real-building use, although it does not develop a broader trust or governance agenda.
Overall RQ7 contribution strength	Strong	The paper identifies a core generalization problem, connects it to real-building deployment limits, and proposes meta-reinforcement learning as a concrete future direction.

Table S7.42: RQ7 challenge and future-direction scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.																				
Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Al Sayed et al. (2024)	3	3	2	0	3	0	0	3	0	0	0	0	0	3	0	3	3	0	3	0

Table S7.43: RQ7 summary table for Al Sayed et al. (2024), after recoding.	
Item	Summary
RQ7 answer	Al Sayed et al. (2024) answer RQ7 by identifying the main barriers to reinforcement-learning-based HVAC control in real buildings and by proposing a future agenda centered on improving adaptation to real-world variability.
Key challenges	Limited real-building deployment, weak generalization caused by insufficient diversification of exogenous training conditions, expensive repetitive retraining and fine-tuning, non-stationary weather, occupancy and electricity-price conditions, and safety concerns in real deployment.
Key future directions	Better training design for real-world variability, more realistic virtual environments, safer simulation-to-real deployment, lower retraining cost, and especially meta-reinforcement learning as a promising route to stronger adaptation.
Overall RQ7 profile	Mostly technical with a clear implementation layer and strong generalization/safety framing; especially useful for showing how later surveys increasingly center the exogenous-variability and generalization problem as a key barrier to real reinforcement-learning deployment in HVAC control.

Table S7.44: One-line RQ7 answer for Al Sayed et al. (2024), after recoding.	
One-line RQ7 answer	
Al Sayed et al. (2024) frame the main challenges of reinforcement-learning-based HVAC control as limited real-building deployment, weak generalization caused by insufficient diversification of exogenous training conditions, and the resulting need for repetitive and computationally expensive retraining, while pointing future work toward more realistic training setups and especially meta-reinforcement learning as a route to safer and more adaptive real-world control.	

S7.12 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [12]

Table S7.45: RQ7 framing and shift coding for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-operational hybrid	The paper frames the main obstacles around sample inefficiency, high-dimensional state/action complexity, difficult reward design, multi-objective trade-offs, weak transferability, interoperability, scalability, safety, privacy, cybersecurity, user trust, and the practical validation of SHEMS with energy storage.
Secondary challenge family	Safety / trust / governance	A strong secondary concern is the practical deployment of smart-home energy management systems with energy storage, including privacy, cybersecurity, interpretability, reliability, policy implications, user acceptance, and real-world testing.
Safety / trust emphasis	Strong	The paper explicitly raises safety, robustness, interpretability, privacy, cybersecurity, policy implications, and user-trust issues as major concerns for real-world implementation.
Dominant future orientation	Scalable, secure, user-aware, and validated reinforcement learning for smart homes with storage	The future agenda is led by hybrid and multi-agent reinforcement learning, stronger demand-side management and demand-response integration, multi-source energy and storage integration, scalability-oriented algorithm design, privacy/security protection, user-preference integration, and more real-world testing and validation.
Shift signal: simulation → real-world deployment	1	The paper explicitly calls for field trials, real-world testing, and validation to bridge the gap between theory and practice.
Shift signal: single-building → community / larger-scale coordination	1	The paper points toward smart-home communities, coordinated devices and storage units, and multi-agent control with centralized training and decentralized execution.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	The paper discusses user acceptance, user preferences, personalized control, privacy, cybersecurity, and interactive reinforcement-learning-based methods.
Shift signal: performance → safety / trust / governance	1	Privacy, cybersecurity, interpretability, policy implications, user trust, and safety are explicitly discussed as part of the field's future needs.
Overall RQ7 contribution strength	Strong	The paper explicitly discusses current limitations, policy and implementation implications, and a detailed multi-part future research agenda.

Table S7.46: RQ7 challenge and future-direction scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Pinthurat et al. (2024)	3	3	3	3	3	3	3	3	3	2	0	1	2	0	3	3	3	3	3	3

Table S7.47: RQ7 summary table for Pinthurat et al. (2024), after recoding.

Item	Summary
RQ7 answer	Pinthurat et al. (2024) answer RQ7 by identifying the main technical, operational, safety, privacy, and implementation barriers of reinforcement-learning-based smart-home energy management with storage and by outlining a broad future agenda aimed at making these systems more scalable, secure, interpretable, user-aware, and practically validated.
Key challenges	Sample inefficiency, high-dimensional state/action spaces, difficult reward design, weak transferability, multi-objective trade-offs, safety and robustness concerns, lack of interpretability, interoperability and scalability problems, cybersecurity/privacy issues, user trust and acceptance, and the practical difficulty of validating systems in real homes.
Key future directions	Hybrid reinforcement learning, multi-agent reinforcement learning with stronger coordination, improved scalability, tighter demand-side management and demand-response integration, privacy and security protection, integration of multiple storage and energy sources, user-preference and personalized control, and substantially more real-world testing and validation.
Overall RQ7 profile	Strongly technical but with a pronounced operational, user-centered, safety, privacy, and policy strand; especially useful for showing how later smart-home surveys move beyond core algorithmic issues toward deployment, interoperability, user trust, cybersecurity, and policy-relevant concerns.

Table S7.48: One-line RQ7 answer for Pinthurat et al. (2024), after recoding.

One-line RQ7 answer
Pinthurat et al. (2024) frame the main challenges of reinforcement-learning-based smart-home energy management with storage as sample inefficiency, high-dimensional and multi-objective control complexity, weak transferability, safety, interpretability, interoperability, scalability, user acceptance, and privacy/cybersecurity concerns, while pointing future work toward hybrid and multi-agent reinforcement learning, stronger demand-side management and demand-response integration, multi-source storage integration, user-aware control, privacy/security protection, and much more real-world testing and validation.

S7.13 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [13]

Table S7.49: RQ7 framing and shift coding for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.		
Dimension	Code / value	Interpretation
Dominant challenge family	Technical-operational hybrid	The paper frames the main obstacles around the complexity of modern interconnected HEMS, high-dimensional and multi-objective decision spaces, large data demands, scalability, interoperability, real-world validation, and privacy/security risks.
Secondary challenge family	Safety / trust / infrastructure	A strong secondary concern is the gap between simulation and practice, including limited empirical real-world studies, interoperability with existing infrastructure, economic viability, cybersecurity, privacy, and deployment at scale.
Safety / trust emphasis	Strong	The paper explicitly raises privacy, cybersecurity, data protection, user trust, and system security as major concerns for practical HEMS deployment.
Dominant future orientation	Scalable, secure, and deployable DRL for HEMS and prosumers	The future agenda is led by stronger data/privacy architectures, scalable cloud/edge-integrated learning, real-world case studies, smart-grid and demand-side response integration, multi-agent or distributed control, and more practical DRL deployment in homes and prosumer buildings.
Shift signal: simulation → real-world deployment	1	The paper explicitly says there is a lack of empirical real-world studies and recommends more practical applications and case studies.
Shift signal: single-building → community / larger-scale coordination	1	The paper extends the discussion toward prosumers, local microgrids, distributed automation, and coordinated control opportunities.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	The paper repeatedly includes user preferences, user satisfaction, privacy-preserving operation, and user trust as important future concerns.
Shift signal: performance → safety / trust / governance	1	Privacy, cybersecurity, compliance, and user trust are explicit parts of the challenge and future-direction framing.
Overall RQ7 contribution strength	Strong	The paper contains explicit sections on problems, gaps, challenges, opportunities, and future perspectives, and ties them to practical HEMS deployment.

Table S7.50: RQ7 challenge and future-direction scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.																				
Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Latoń et al. (2024)	3	2	3	3	3	3	3	2	3	1	0	1	2	0	3	3	3	2	2	3

Table S7.51: RQ7 summary table for Latoń et al. (2024), after recoding.	
Item	Summary
RQ7 answer	Latoń et al. (2024) answer RQ7 by identifying the main barriers to RL/DRL-based HEMS as complexity, scalability, interoperability, privacy/security, data demands, and weak real-world validation, and by outlining a future agenda aimed at making HEMS more scalable, secure, interoperable, and practically deployable.
Key challenges	High-dimensional and multi-objective control complexity, limited empirical real-world studies, scalability and interoperability problems, privacy and cybersecurity risks, dependence on large and reliable datasets, changing market/weather/user-preference conditions, and unresolved practical integration with existing infrastructure.
Key future directions	More practical deployments and case studies, stronger data and security architectures, scalable cloud/edge/distributed learning frameworks, greater smart-grid and demand-side response integration, multi-agent coordination, adaptability to changing operating conditions, and stronger prosumer or local-microgrid applications.
Overall RQ7 profile	Strongly technical with a pronounced implementation and privacy/security strand; especially useful for showing how later HEMS reviews move beyond algorithmic performance toward deployment, interoperability, user trust, and infrastructure-level readiness.

Table S7.52: One-line RQ7 answer for Latoń et al. (2024), after recoding.	
One-line RQ7 answer	
Latoń et al. (2024) frame the main challenges of RL/DRL-based HEMS as the complexity of modern interconnected home-energy systems, limited real-world validation, scalability and interoperability barriers, and privacy/security risks, while pointing future work toward more practical case studies, stronger data and security frameworks, multi-agent and smart-grid-integrated control, and scalable deployment in prosumer and local-microgrid settings.	

S7.14 Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [14]

Table S7.53: RQ7 framing and shift coding for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-human-centric hybrid	The paper frames the main obstacles around multi-time-scale control, complex dynamics, occupant behavior, high-dimensional multi-objective control, heterogeneity and scalability, interpretability/transparency, computational burden, and high data requirements.
Secondary challenge family	Operational / reproducibility / safety	A secondary concern is the need for open data, open-source code, better reproducibility practices, secure and transparent decision-making, and more practically deployable MORL systems across varied building contexts.
Safety / trust emphasis	Strong	The paper explicitly raises interpretability, transparency, security, access, safety management, stakeholder trust, and robustness against unseen uncertainties as important concerns for real-world MORL adoption.
Dominant future orientation	Human-centric, scalable, secure, and transferable MORL for smart buildings	The future agenda is led by incremental and transfer learning, dynamic human-centric reward design, hybrid IoT-enabled models, renewable and storage integration, decentralized multi-agent systems, and stronger security, transparency, and safety management.
Shift signal: simulation → real-world deployment	1	The paper partially signals this shift by emphasizing practical deployment, robustness, reproducibility, and real-world smart-building applicability, even though much of the literature remains method-focused.
Shift signal: single-building → community / larger-scale coordination	1	The paper points toward decentralized and multi-agent systems, shared flexibility, and broader coordination across building systems and zones.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	Occupant behavior, human-centric rewards, comfort preferences, and user adoption costs are explicitly emphasized.
Shift signal: performance → safety / trust / governance	1	Interpretability, transparency, security, access, and safety management are directly identified as future priorities.
Overall RQ7 contribution strength	Strong	The paper has an explicit challenge section, an explicit future-directions section, and a conclusion that consolidates both into a coherent MORL research agenda.

Table S7.54: RQ7 challenge and future-direction scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Huynh et al. (2025)	2	3	3	3	3	3	2	3	3	1	0	3	2	0	3	3	2	3	3	3

Table S7.55: RQ7 summary table for Huynh et al. (2025), after recoding.

Item	Summary
RQ7 answer	Huynh et al. (2025) answer RQ7 by identifying the main challenges of MORL in smart buildings as multi-time-scale control, complex dynamics, occupant behavior, heterogeneity and scalability, interpretability/transparency, computational burden, data quality, and high data requirements, while outlining a broad future agenda for more scalable, human-centric, secure, and transferable MORL systems.
Key challenges	Multi-time scales, high-dimensional and multi-objective control complexity, complex dynamics, occupant-behavior uncertainty, heterogeneity and scalability across buildings and zones, interpretability/transparency, security and safety management, computational complexity, training-resource burden, and the need for large amounts of high-quality data.
Key future directions	Incremental and transfer learning, dynamic human-centric reward structures, hybrid models with IoT devices, renewable and storage integration, decentralized and multi-agent systems, better space and equipment utilization, robustness to unseen uncertainty, and stronger security, access, and safety management.
Overall RQ7 profile	Strongly technical, but with a clear human-centered, operational, reproducibility, and safety/transparency strand; especially useful for showing how later MORL research broadens from pure optimization toward transferability, occupant-centric control, reproducibility, and secure deployment.

Table S7.56: One-line RQ7 answer for Huynh et al. (2025), after recoding.

One-line RQ7 answer
Huynh et al. (2025) frame the main challenges of MORL in smart buildings as multi-time-scale and high-dimensional control complexity, occupant uncertainty, heterogeneity and scalability, interpretability/transparency, computational/data burden, and security/safety concerns, while pointing future work toward transfer and incremental learning, human-centric reward design, hybrid IoT-enabled models, decentralized multi-agent systems, renewable/storage integration, and stronger transparency, security, and safety management.

S7.15 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [15]

Table S7.57: RQ7 framing and shift coding for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Operational / implementation-dominant	The paper frames the main obstacles around deploying reinforcement learning safely and robustly in real buildings, scaling solutions beyond tailored pilot cases, obtaining long-term whole-system evidence, and building trustworthy digital-twin-supported deployment workflows.
Secondary challenge family	Technical / generalization	A secondary concern is that pretraining is often tied to customized simulation models, which limits transferability and raises questions about how well controllers generalize across buildings, retrofits, and changing occupant behavior.
Safety / trust emphasis	Strong	Safety, security, privacy, transparency, human oversight, fairness, and accountability are not peripheral issues here; they are treated as core barriers to trustworthy real-building adoption.
Dominant future orientation	Digital-twin-supported, trustworthy, and scalable real-building RL deployment	The future agenda is led by stronger digital-twin support for development and operation, better transferability and long-term validation, and explicit human-centered, privacy-aware, transparent, and accountable AI design.
Shift signal: simulation → real-world deployment	1	The paper is explicitly about implementation hurdles in real buildings and repeatedly stresses the need to move beyond short, limited deployments.
Shift signal: single-building → community / larger-scale coordination	0	This is not a main organizing shift in the paper, even though data-rich digital twins could support coordination across interacting systems.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	The paper explicitly elevates human-centricity, occupant comfort, user trust, privacy, fairness, and human oversight as essential parts of future building AI deployment.
Shift signal: performance → safety / trust / governance	1	This is one of the clearest themes of the paper, with safety, security, privacy, transparency, governance, accountability, and ethical oversight treated as major future priorities.
Overall RQ7 contribution strength	Strong	The paper is explicitly organized around implementation challenges, digital-twin deployment support, and ethical/governance requirements for real-building AI control.

Table S7.58: RQ7 challenge and future-direction scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Kannari et al. (2025)	3	2	1	0	3	1	3	3	3	0	0	3	0	0	1	3	3	3	3	3

Table S7.59: RQ7 summary table for Kannari et al. (2025), after recoding.

Item	Summary
RQ7 answer	Kannari et al. (2025) answer RQ7 by identifying the main barriers to real-building reinforcement-learning adoption as safe deployment of adaptive controllers, weak scalability and transferability beyond tailored pilot cases, limited long-term and whole-system evidence, and the lack of systematic ethical analysis in reported implementations.
Key challenges	Short and narrow deployment studies, dependence on customized simulation pretraining, uncertainty about long-term robustness under retrofits and changing occupant habits, practical preprocessing and safety-check needs, limited whole-building evidence, digital-twin maturity, and underdeveloped treatment of privacy, transparency, human oversight, fairness, and accountability.
Key future directions	Better transferability across buildings, clearer maturity criteria and validation for digital twins, more long-term and holistic real-building studies, stronger digital-twin-supported deployment workflows, and explicit ethical design based on human-centricity, safety/security, privacy/data protection, transparency, fairness, and governance/accountability.
Overall RQ7 profile	Strongly implementation-centered, with a pronounced safety, ethics, and governance strand; especially useful for showing how the field’s priorities are shifting from technical promise in simulation toward trustworthy, explainable, and scalable deployment in real buildings.

Table S7.60: One-line RQ7 answer for Kannari et al. (2025), after recoding.

One-line RQ7 answer
Kannari et al. (2025) frame the main challenges of reinforcement-learning-based control in real buildings as safe deployment, weak scalability and transferability, limited long-term whole-system evidence, digital-twin maturity, and insufficient ethical/governance treatment, while pointing future work toward digital-twin-supported deployment, stronger validation and transfer practices, and explicitly human-centered, transparent, privacy-aware, and accountable AI control.

S7.16 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [16]

Table S7.61: RQ7 framing and shift coding for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Dimension	Code / value	Interpretation
Dominant challenge family	Technical-dominant	The paper frames the main obstacles around the complexity of RES-integrated BEMS, difficult multi-objective control, the limits of simple value-based reinforcement learning in high-dimensional settings, transfer across buildings and climates, and the coordination burden of multi-agent and hybrid energy systems.
Secondary challenge family	Operational / implementation	A strong secondary concern is the transition from simulation to practice, including limited real-world experimentation, real-time implementation demands, safety/compliance needs, and the need to coordinate diverse assets under practical constraints.
Safety / trust emphasis	Moderate–strong	Safety and compliance are explicit future priorities, especially through constrained and shielded reinforcement learning, although broader ethics and governance themes are less developed than the technical safety agenda.
Dominant future orientation	Scalable, safe, transferable, and coordinated reinforcement learning for RES-integrated BEMS	The future agenda is led by actor–critic and hybrid reinforcement learning, hierarchical and multi-agent coordination, stronger transfer and adaptation across buildings and climates, safer control under constraints, and broader real-world deployment in complex RES-integrated systems.
Shift signal: simulation → real-world deployment	1	The paper explicitly notes that real-life experiments are still rare and that future work must prioritize practical implementation with fallback and safety mechanisms.
Shift signal: single-building → community / larger-scale coordination	1	The paper points toward microgrids, districts, hierarchical multi-agent reinforcement learning, and coordination of multiple interacting subsystems.
Shift signal: pure optimization → human-centered / stakeholder-centered framing	1	Occupant comfort remains a core objective and future systems are framed as needing to balance technical optimization with occupant-centric constraints.
Shift signal: performance → safety / trust / governance	1	Safety and regulatory compliance become explicit future priorities, alongside privacy-preserving federated approaches.
Overall RQ7 contribution strength	Strong	The paper explicitly structures its discussion around current trends and future directions, linking algorithmic choices to deployment barriers, coordination needs, safety, transfer, and system-scale complexity.

Table S7.62: RQ7 challenge and future-direction scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.																				
Paper	C1	C2	C3	C4	C5	C6	C7	C8	C9	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
Michailidis et al. (2025)	3	2	3	3	3	3	2	3	2	2	0	3	2	2	3	3	3	2	3	2

Table S7.63: RQ7 summary table for Michailidis et al. (2025), after recoding.	
Item	Summary
RQ7 answer	Michailidis et al. (2025) answer RQ7 by identifying the main barriers to reinforcement-learning-based renewable-energy building control as the complexity of RES-integrated BEMS, difficult comfort–cost–grid trade-offs, limits of simple RL methods in continuous high-dimensional settings, coordination burdens in multi-agent systems, weak transfer across buildings and climates, and the still-limited move from simulation to real deployment.
Key challenges	High-dimensional and continuous control complexity, competing objectives across comfort, cost, emissions, and grid services, communication and coordination overhead in multi-agent reinforcement learning, poor transferability across building types and climates, safety and compliance needs, privacy concerns, and sparse real-world experimentation.
Key future directions	More actor–critic and hybrid reinforcement learning, hierarchical and scalable multi-agent reinforcement learning, broader use of thermal and non-battery storage resources, stronger transfer/domain adaptation, constrained and shielded reinforcement learning for safety/compliance, privacy-preserving distributed learning, and substantially more real-world deployment.
Overall RQ7 profile	Strongly technical with a clear implementation and safety layer; especially useful for showing how reinforcement-learning priorities in renewable-energy-integrated buildings are shifting from simple discrete optimization toward coordinated, safe, transferable, and real-time control of multiple interacting assets.

Table S7.64: One-line RQ7 answer for Michailidis et al. (2025), after recoding.	
One-line RQ7 answer	
Michailidis et al. (2025) frame the main challenges of reinforcement-learning-based renewable-energy building control as high-dimensional multi-objective complexity, multi-agent coordination overhead, weak transfer across buildings and climates, safety/compliance constraints, privacy concerns, and lack of real-world deployment, while pointing future work toward actor–critic and hybrid reinforcement learning, hierarchical multi-agent coordination, broader storage integration, transfer/domain adaptation, constrained safe reinforcement learning, and wider practical implementation in RES-integrated BEMS.	

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